GRASP Builder User Manual

Master Thesis
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by

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Advisor: name of the advisor
Barcelona, Date XXXXX

Contents

1 Set up

GRASP Builder application can be executed in both operative systems, Windows and Linux. The only prerequisite in order to work with this application is to have MATLAB installed.

The only functionality not enabled in Windows is the execution of GRASP algorithm, since this application is desigend to work in CALCULA enviroment, but all the other funcionalities can be used, if the user prefers.

1.1 Windows

Since this application is not being installed with any executable or saved in registry, it is only necessary to unzip file *GRASP_Builder_v200_Windows.zip* in the desired folder.

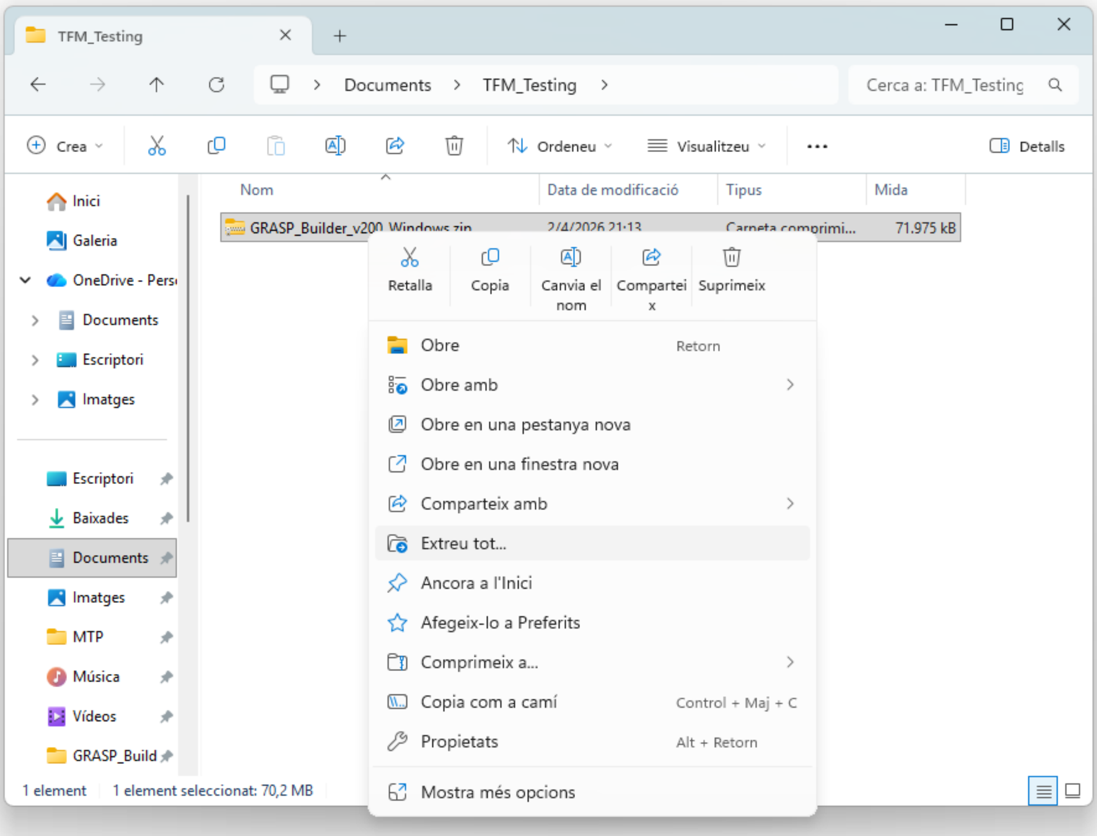

Figure 1: Unzip process in Windows

In the unziped folder it will be located and executable called GRASP_Builder.exe and it can be executed.

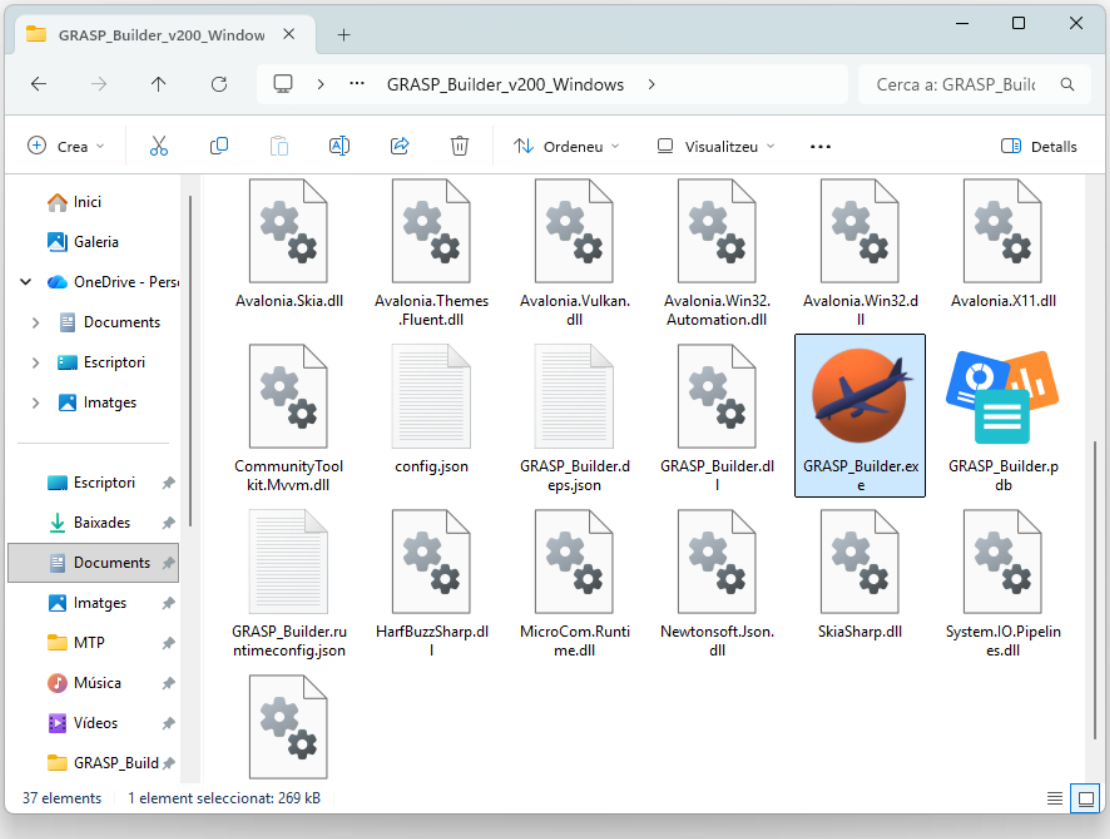

Figure 2: Executable in Windows

1.2 Linux: CALCULA

Similar procedure needs to be followed in the Linux enviroment, but it all can be executed via command line. In this case, it is needed an extra step, since Linux needs to have specified and application as executable.

In the desired folder, it is needed to open a command line. It can be done as it is shown in the following image:

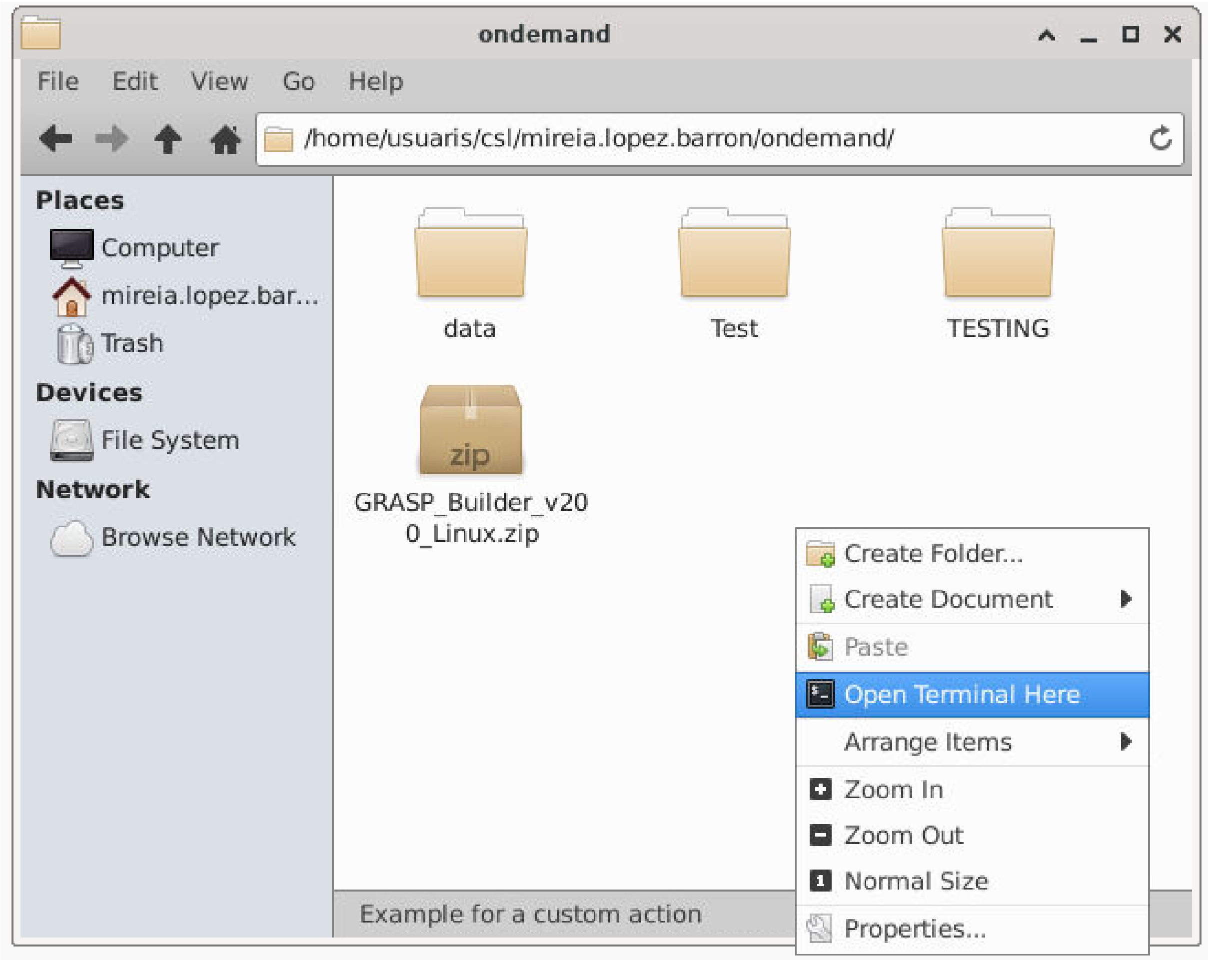

Figure 3: Open cmd in desired folder using Linux

Once it is opened, the following lines have to be executed, in order to unzip files, move to new folder and give executable permissions to the application.

```
unzip GRASP_Builder_v200_Linux.zip
cd GRASP_Builder_v200_Linux
chmod +x GRASP_Builder
```

With these commands, the application has been prepared to be opened whenever it is desired by using the command:

```
./GRASP_Builder
```

Since this application is not shortcuted with a command, every time it is desired to be used, it has to be accessed the installation folder via command line and execute the calling command with the application name.

2 Project management

When the application is launched, the home screen is displayed showing three diferent buttons to manage projects.

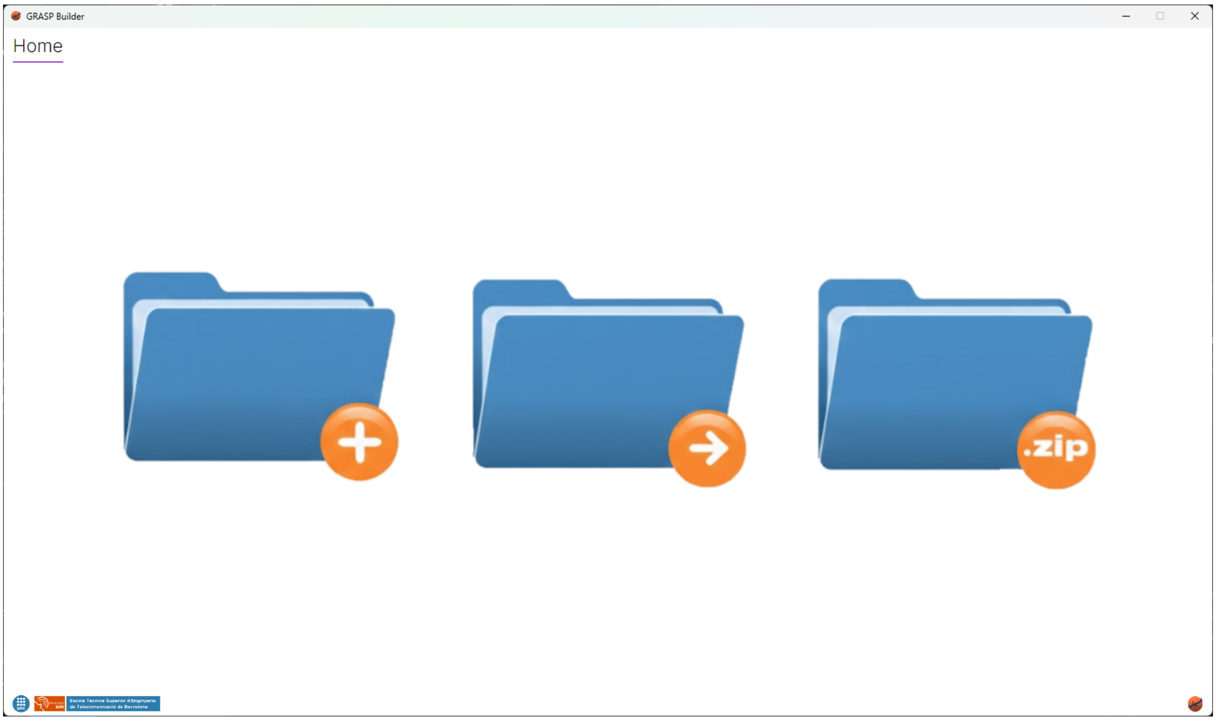

Figure 4: Home screen

First button will open a dialog to create a new project (Figure 5). Second button will open a dialog that allows the user to open an existing project (Figure 6). Third button will open a dialog that allows the user to import a project from a zip file (Figure 9).

When creating a new project, the user must provide a name for the project and select a location to save it. In the project folder, a new file called ”project.grasp” will be created. This file will contain the corresponding project configuration.

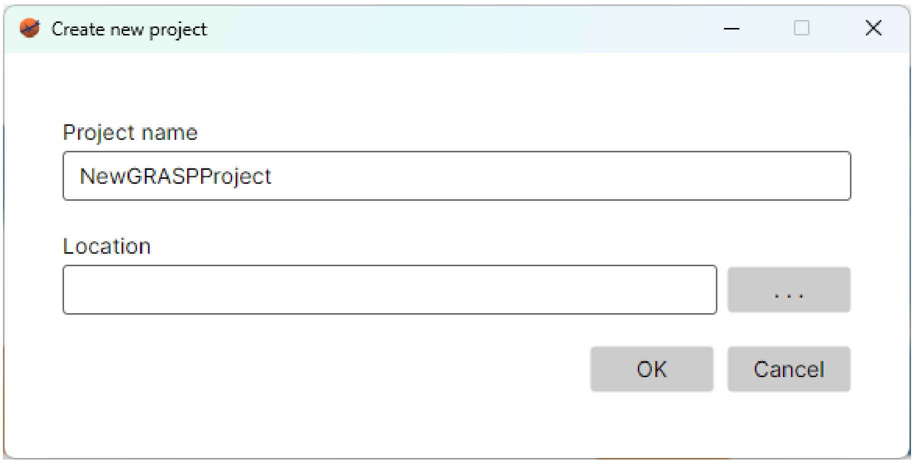

Figure 5: Create new project window

When opening a project, the user must select a project folder that contains an existing project.

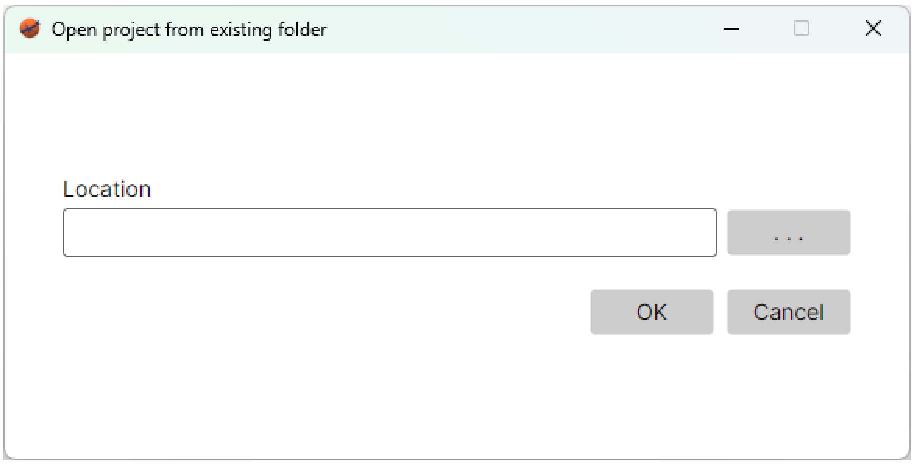

Figure 6: Open project window

All dialogs allow the user to navigate through the file system to select a project folder by using the browser button "..." (Figure 7).

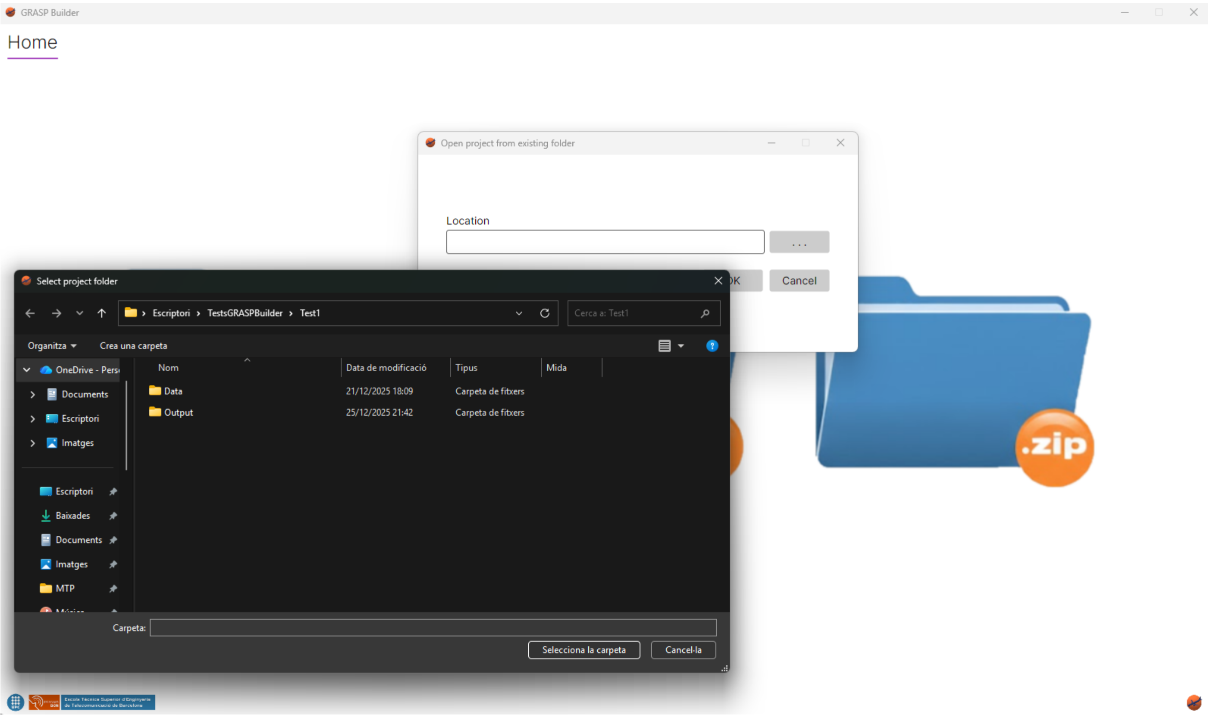

Figure 7: Open project browser

In the project folder, it is necessary it exists the file "project.grasp". Without this file, the application will not be able to load the project and an error message will be displayed (Figure 8).

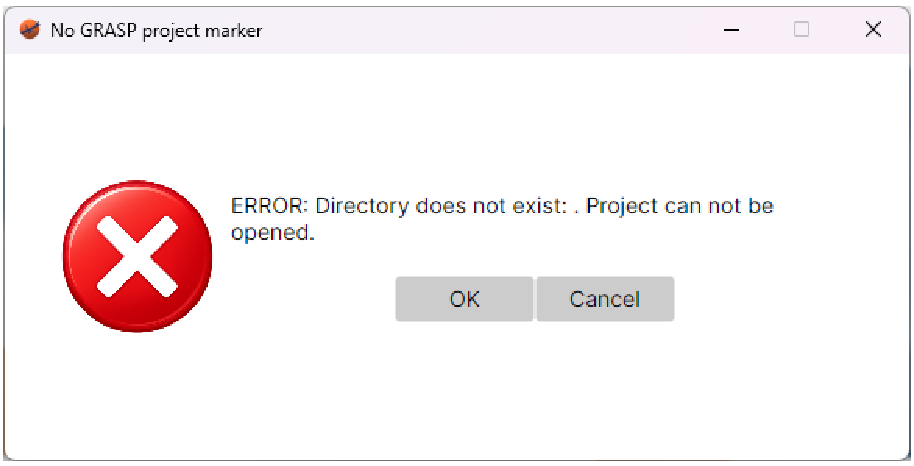

Figure 8: Error when opening a project

When importing a project, the user must select a zip file that contains an existing project. This zip file must also contain the file "project.grasp". Zip file should be located in the directory where the user wants the project to be created.

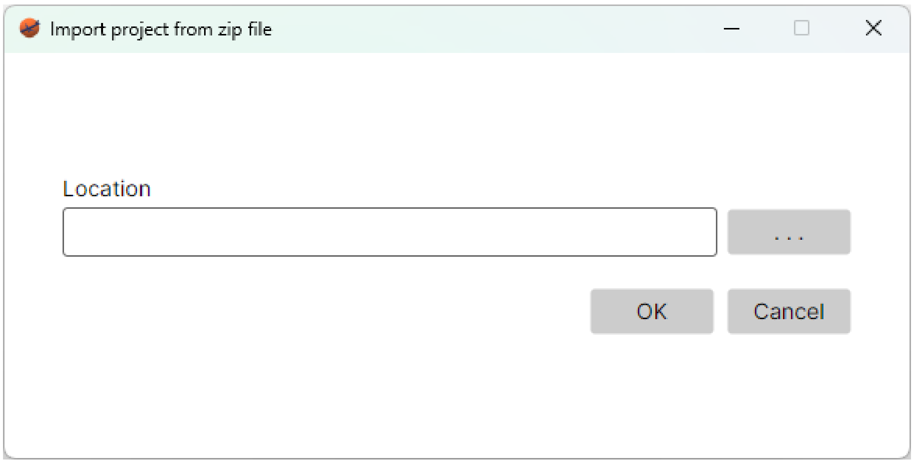

Figure 9: Import project window

Once a project is initialized, the user can access the File menu to perform different actions, in order to manage the current project (Figure 10).

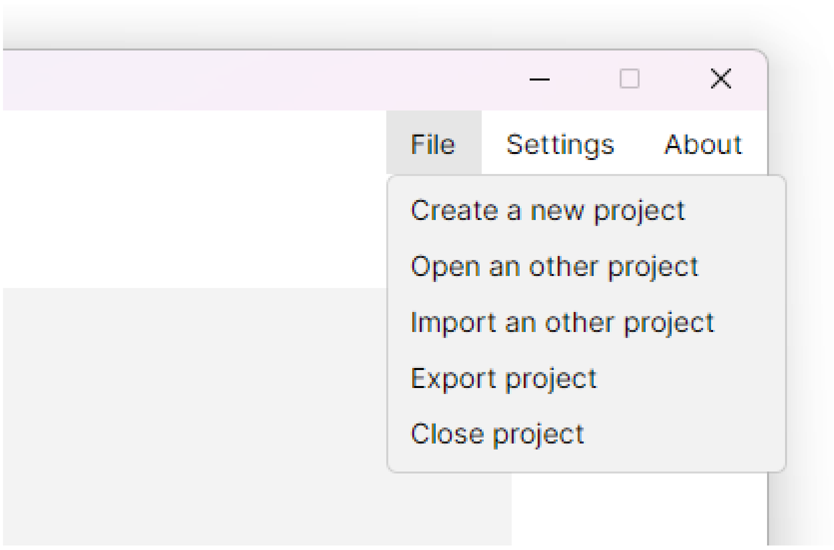

Figure 10: Menu options

Next to the File menu, it can be located the Settings button. This button opens a dialog that allows the user to configure different options of the application Figure 11.

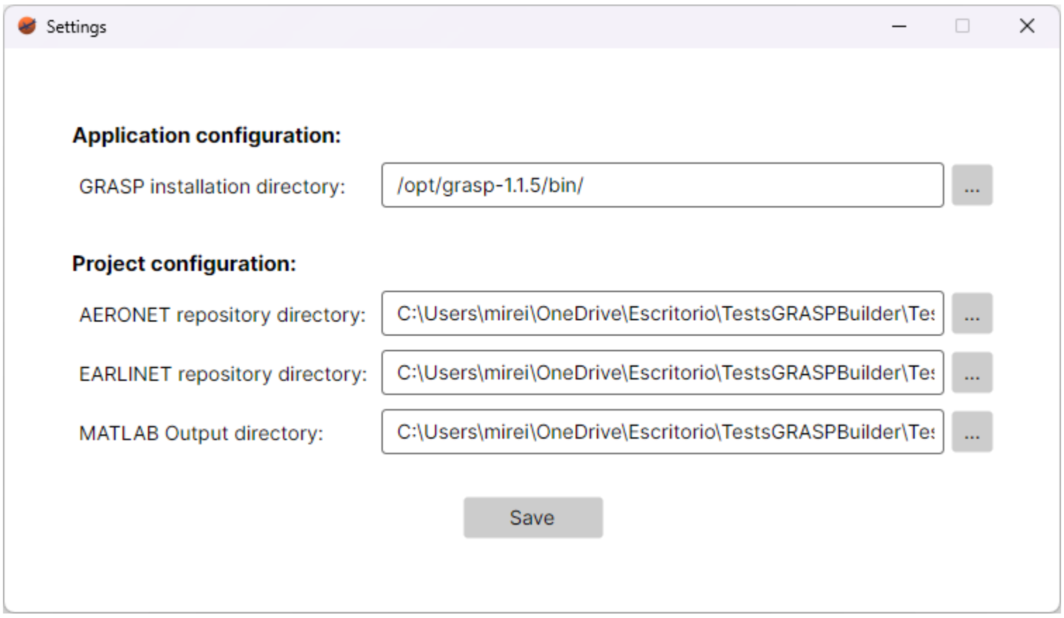

Figure 11: Settings window

The different parameters that can be configured are separated by application configuration parameters and project configuration parameters. The application configuration parameters have the same value for all projects. For now, it only includes the installation directory of GRASP algotithm.

Project configuration parameters are specific for each project. For now, it only includes the corresponding respository directories where the downloaded data will be stored for both, EARLIENT and AERONET sites, and the MATLAB output directory, where all the files created after running the corresponding scripts and GRASP algotithm will be created.

3 Download files

Once a project is created, the first step is to download the files from the different websites in order to have the data available.

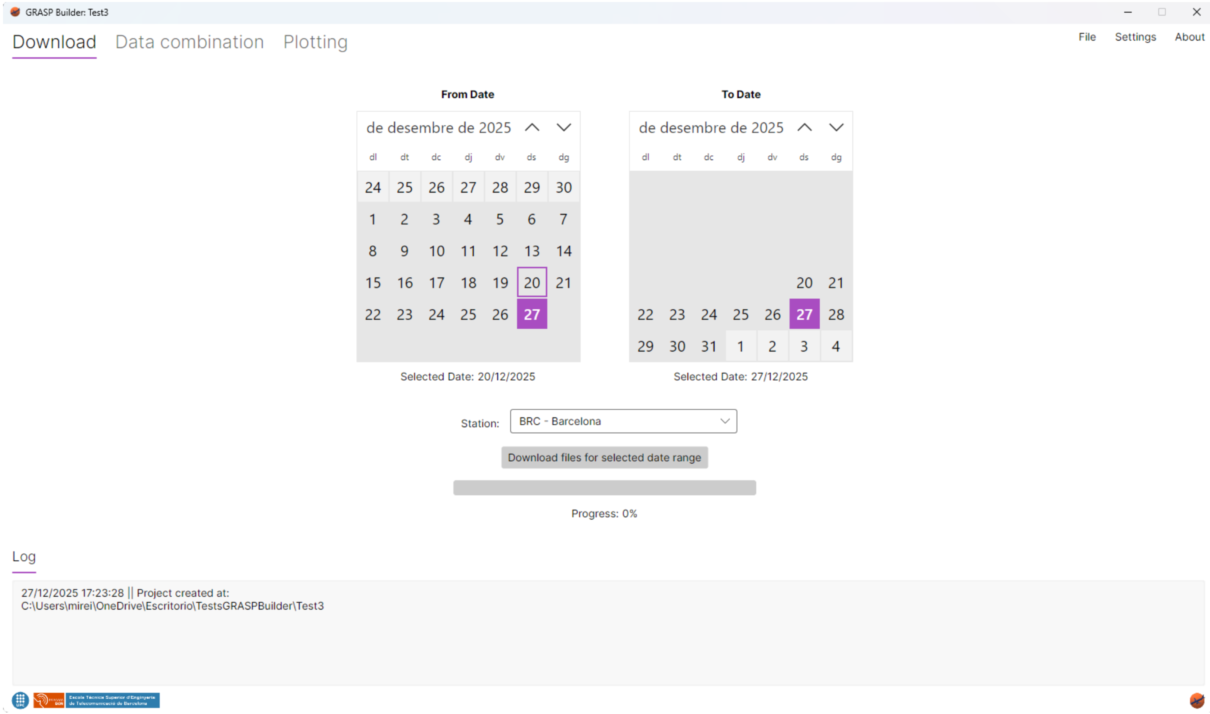

Figure 12: Project loaded

In this tab, the user is available to choose the range of dates to download the files from the different websites. For the moment, it is only available to download files from the selected site. Available sites are locations that have data in both websites, Aeronet and Earlinet. Sites that only have data in one of them are descarted, since it is necessary files from both platforms in order to run the GRASP algorithm.

During the process, all buttons will be disabled and the progress bar will show the percentage of files downloaded. Once all files are downloaded, a new section will be shown where the user will be able to see the list of files downloaded and select the files that desires to keep. This section will be also shown if the user opens a project with already downloaded files (Figure 13).

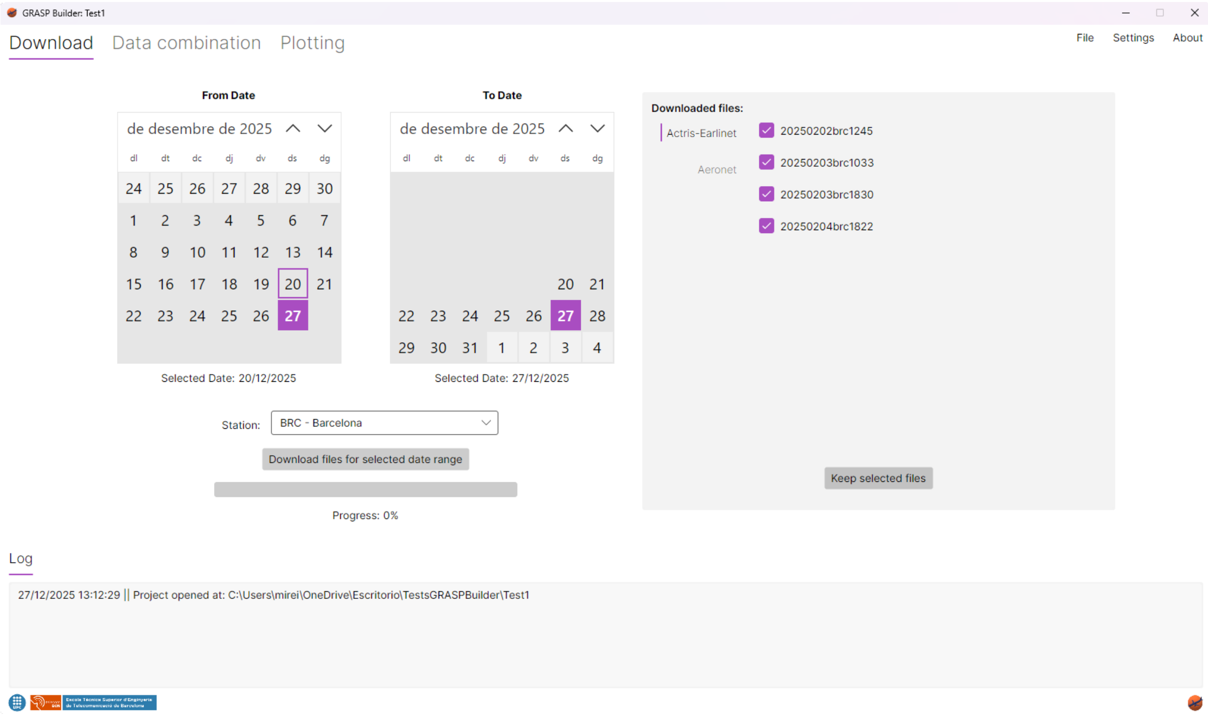

Figure 13: Files downloaded

4 Data Combination

Once all files have been downloaded, the user should the Data Combination tab. In this view, it can be selected a Measure ID from the options available (Figure 14). All measure IDs shown in downloaded files my be not available, since the list of measure IDs is filtered depending on the data contained. Measure IDs containing files with code 007 will not be added to the list, since these measurements will not contain all the data needed for the analysis.

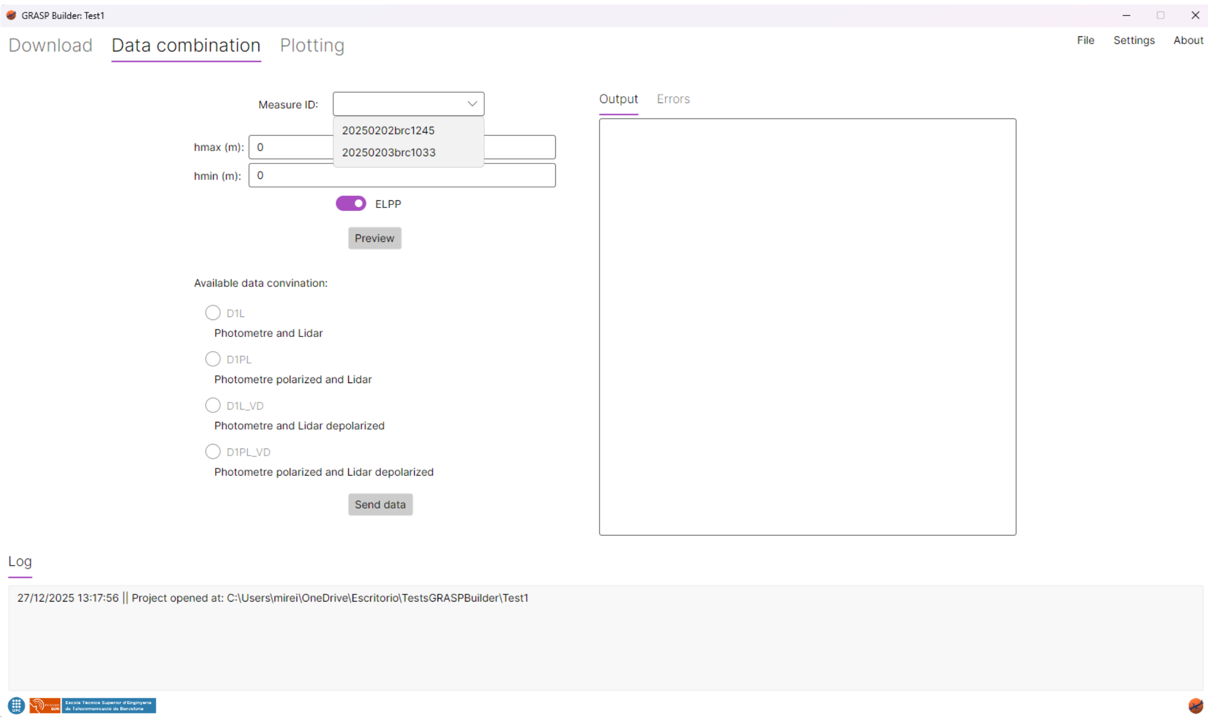

Figure 14: Data combination view

When a Measure ID is selected, the GetHeights script is executed, so the minimum and maximum height can be retrieved instead of having 0 values. The, the Preview script has to be launched in order to preview the data and obtain the available configuration options. This script can be launched manually using the ”Preview” button all the desired times with the two different type of data (ELPP and VD, if possible). Default configured data type to show is ELPP, since this data is always available (Figure 15). The other option is VD (Figure 16).

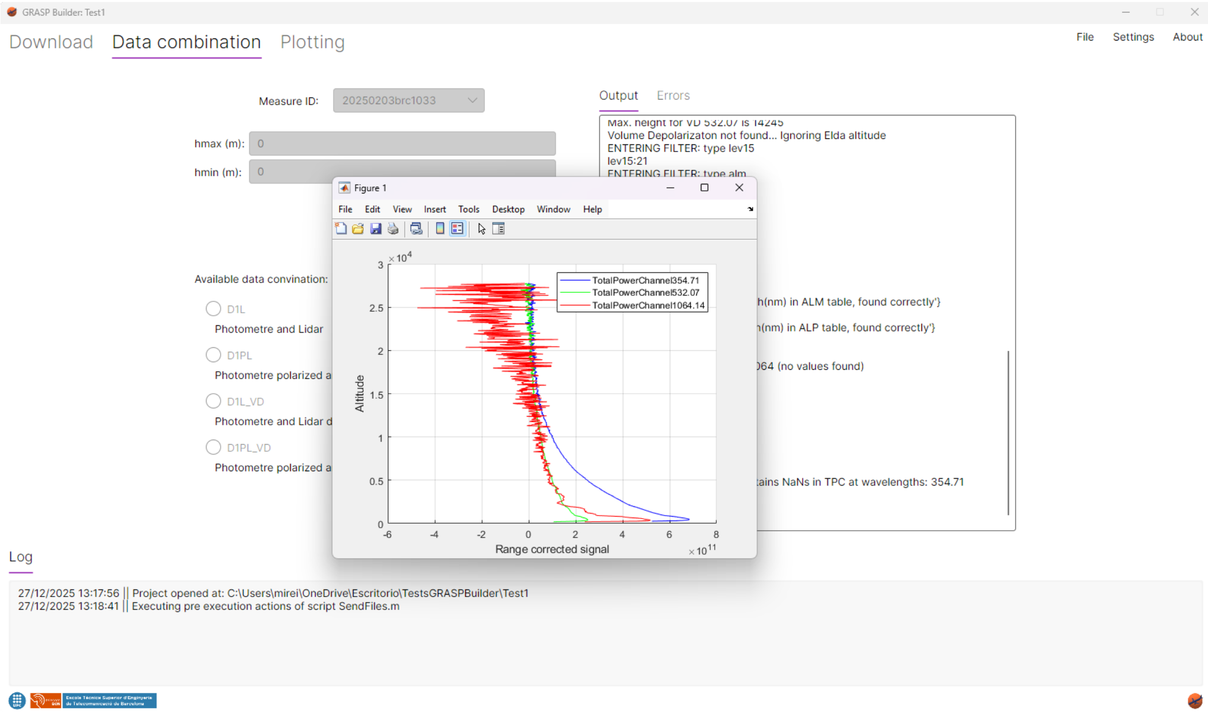

Figure 15: Execute preview with ELPP option

During the script execution, the previsualitzatin figure will be shown. In order to update the application with the available configurations, the Matalb figure needs to be closed, in order to execute the corresponding script post execution actions.

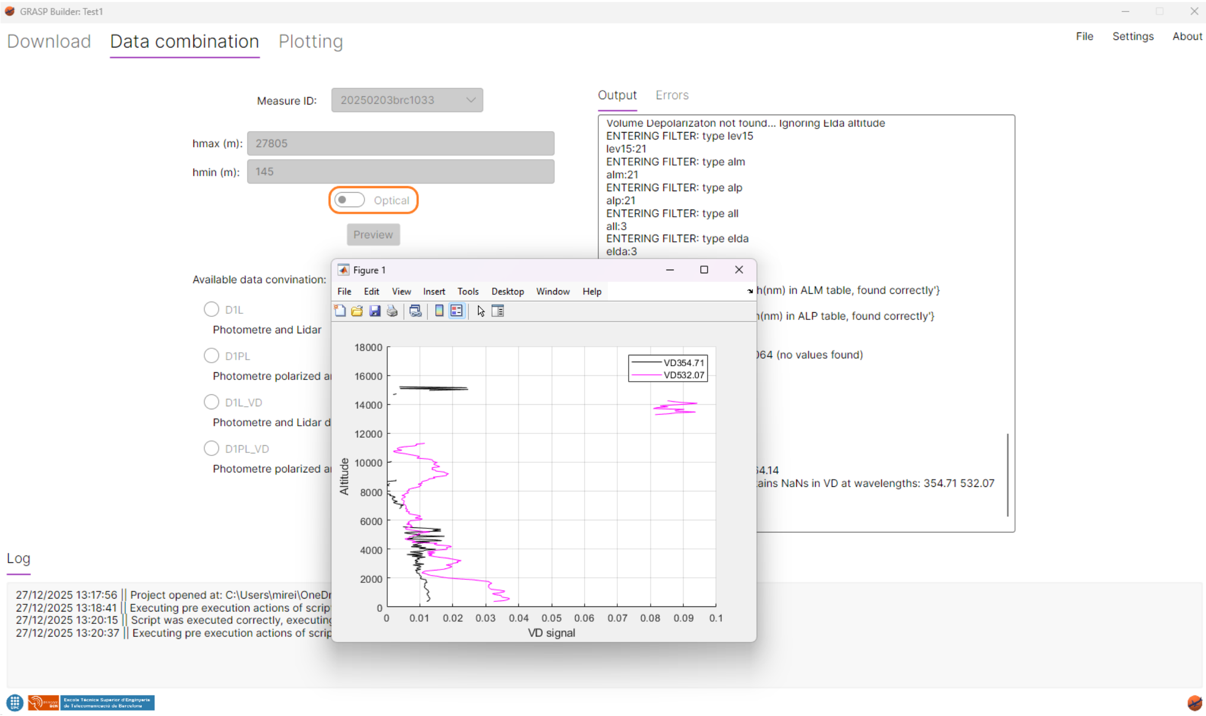

Figure 16: Execute preview with VD option and heighs defined

It is recommended to review the height values before starting the data combination process, since NaN values must be avoided in order to run the algorithm successfully.

Figure 17 shows and example of the preview window with the heights defined and Optical data selected.

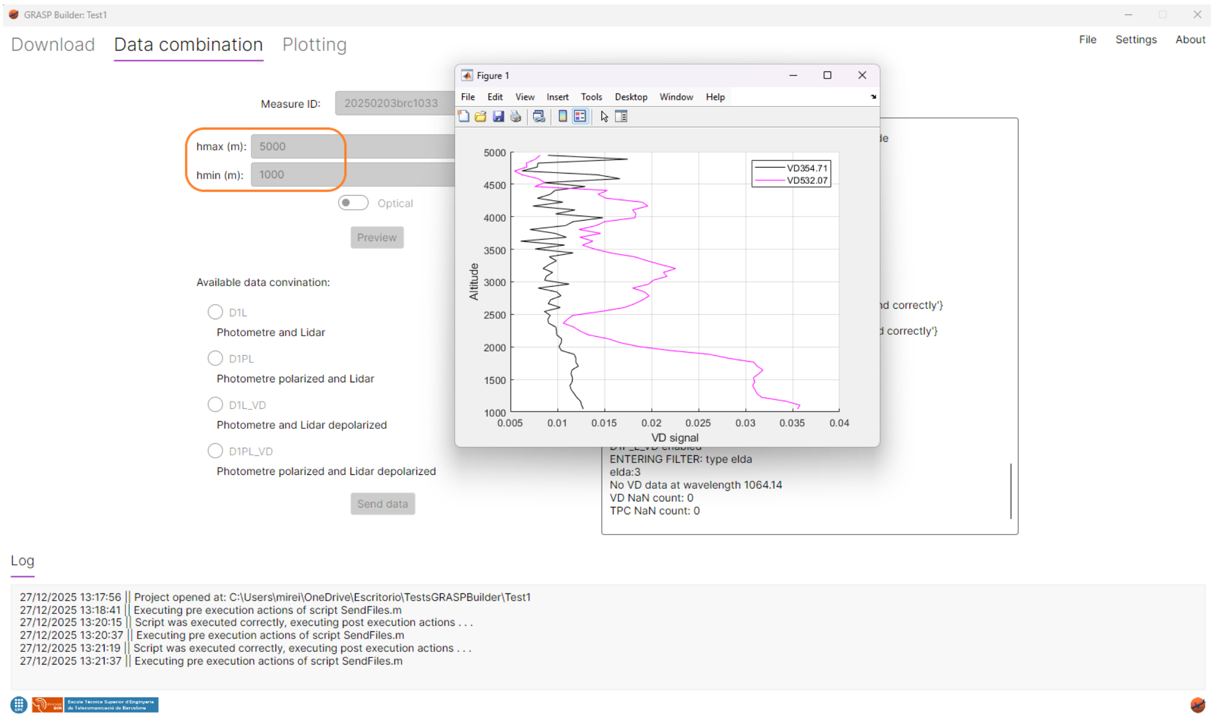

Figure 17: Execute preview with VD option

Every time a script in this window is executed, all buttons will be disabled, and once the plot window is closed, GUI elements will be updated and enabled again.

When a configuration is selected, by clicking the "Send ata" button, the SendFiles script is executed. If the script is executed successfully, the corresponding messages will be shown in Ouput and Logging tabs, as it can be seen in the following example in Figure 18.

If .sdat file and .yml has been written successfully, GRASP algorithm will be executed automatically.

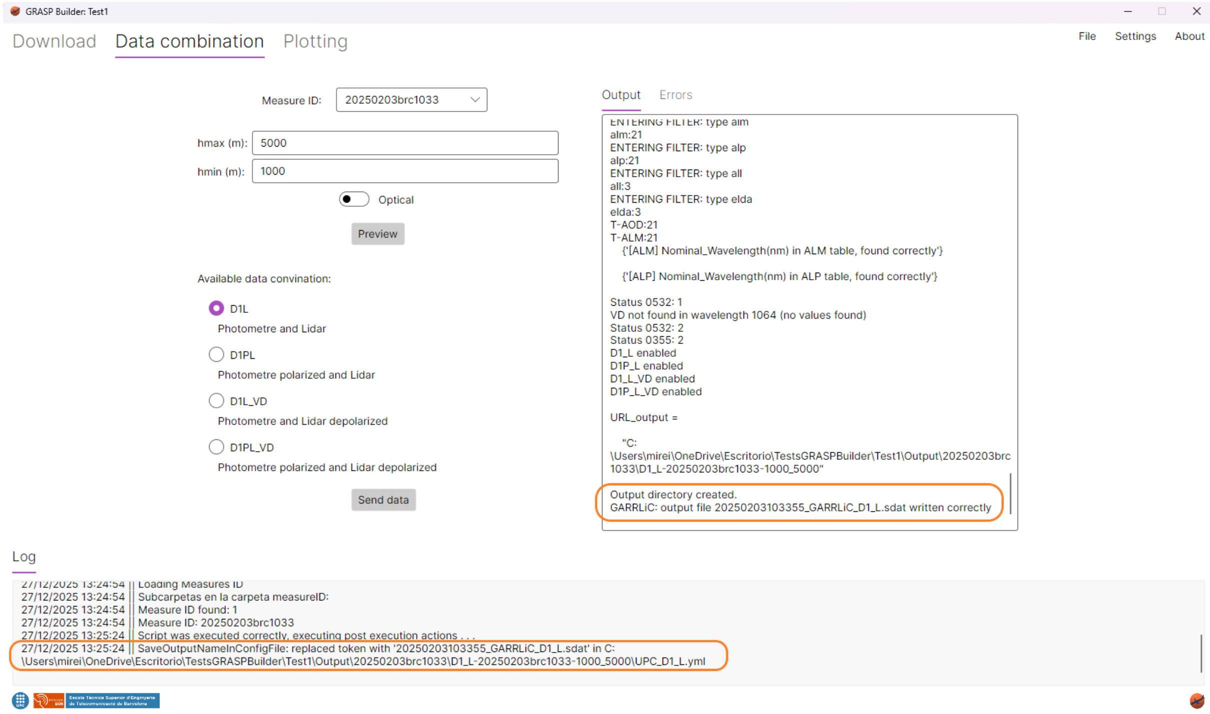

Figure 18: Execute SendFiles script

5 Plotting

Once GRASP has been executed, the user can plot the result data in the Plotting tab (Figure 19).

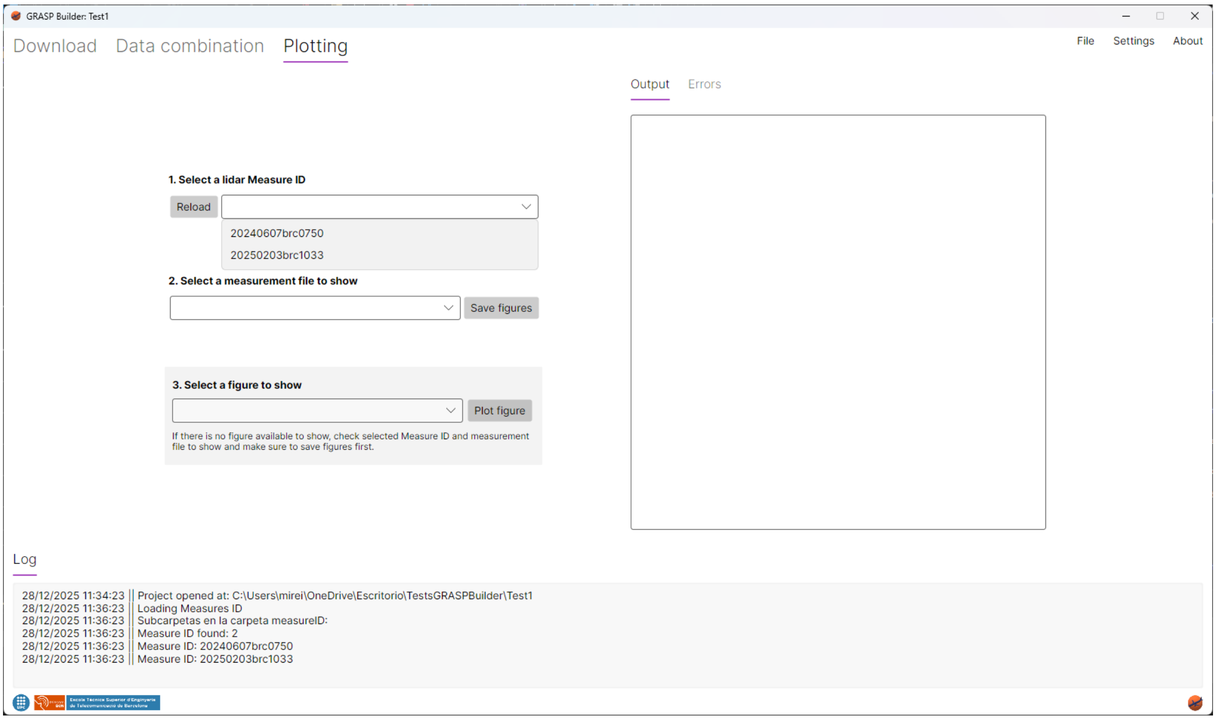

Figure 19: Plotting Tab with available Measure IDs

In this tab, the user can see the available Measure IDs and select the ones that desires to plot. If there are no Measure IDs available, click in "Reload" button, it will scan the MatalabOutputDirectory folder again.

Once the Measure ID is selected, the list of available measurement files will be updated automatically. There the user will be able to choose between the different options (Figure 20).

If this process has already been executed, it will not be needed to execute "Save figures". In the case it is not it is necessary to do it, in order to read all the GRASP output data and save it in the corresponding files.

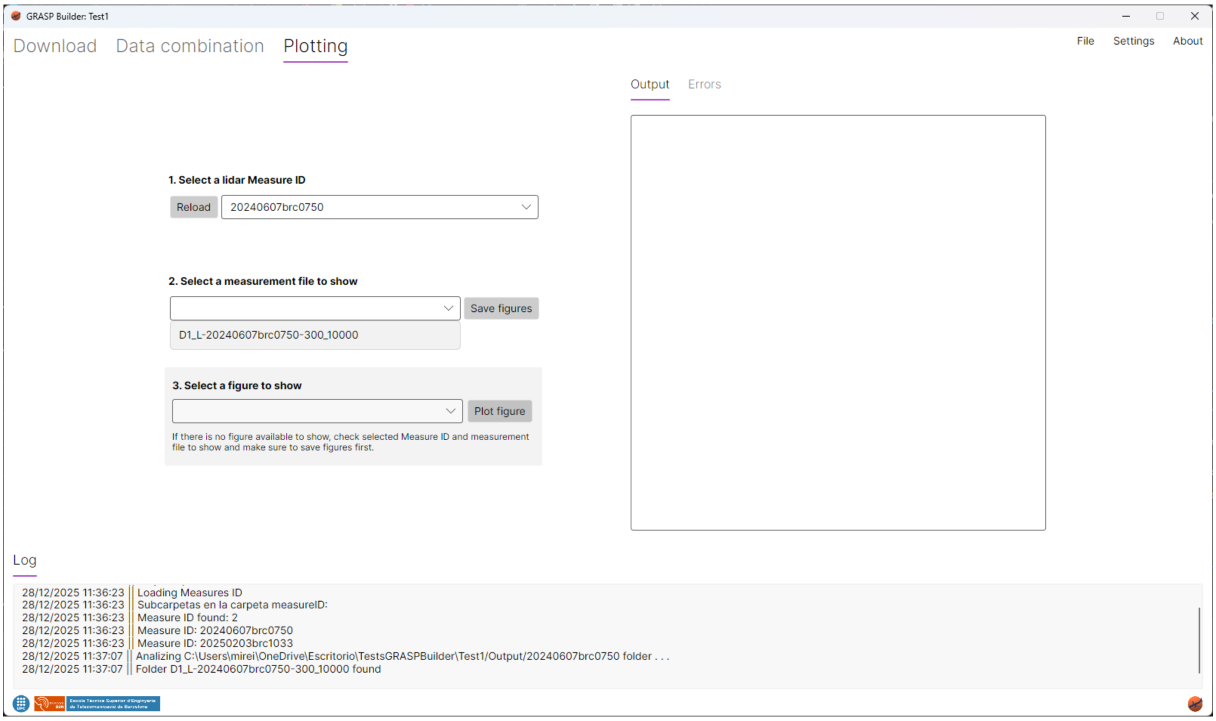

Figure 20: Plotting Tab with available measurement files

All available figures will be listed (Figure 21), the user will be able to choose between the different options.

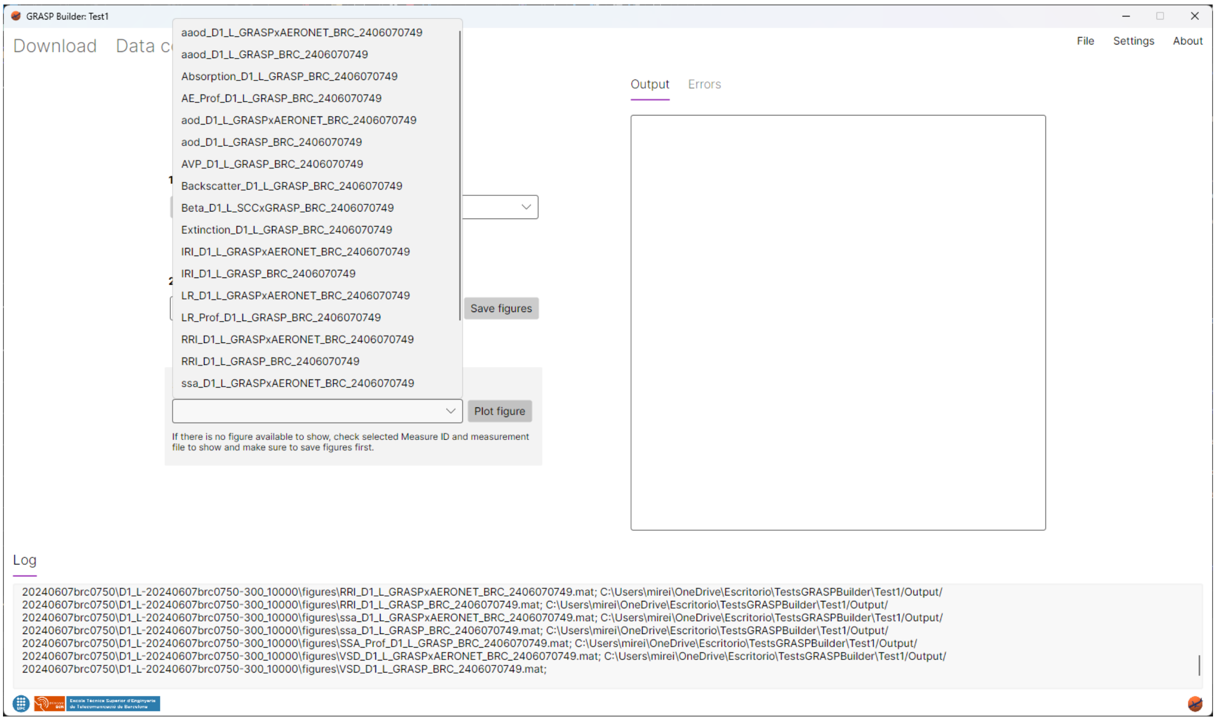

Figure 21: Plotting Tab with available figures

Once a figure is chosen, button "Plot figure" can be clicked in order to start corresponding MATLAB script execution. During the script execution, the corresponding figure will be shown (Figure 22).

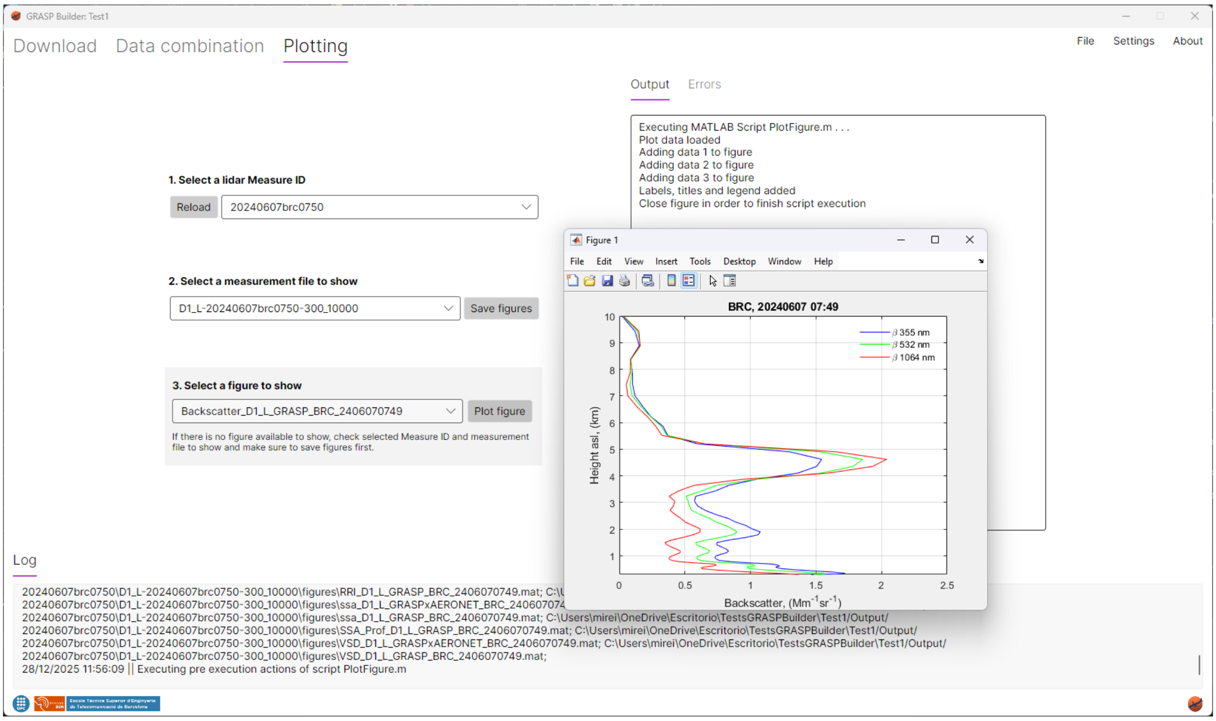

Figure 22: Plotted figure example